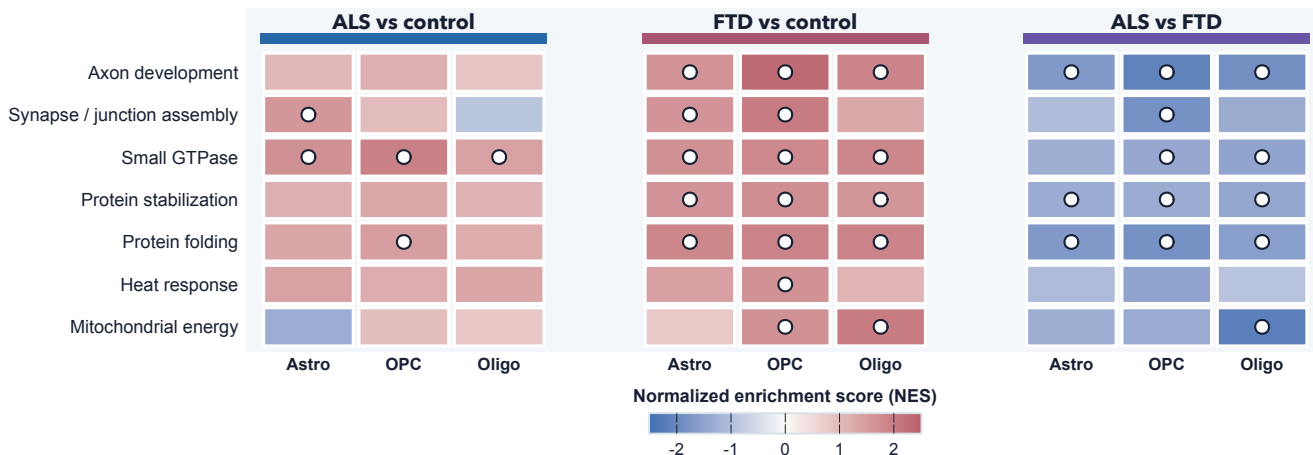


a

Programme effects separate by cell type and phenotype contrast

Seven prespecified mechanism families × three cell types × three contrasts; white rings mark global primary FDR < 0.05 (35/63 axes).

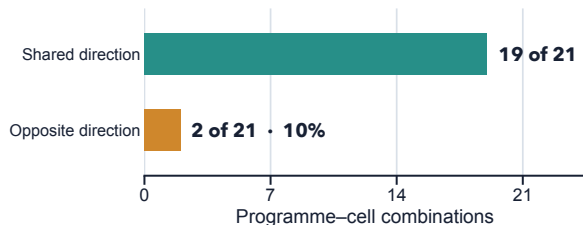


b

Phenotype-direction classes

19 shared, 2 opposite; all direct ALS-versus-FTD contrasts were FTD-higher.

Descriptive classification; no separate test.

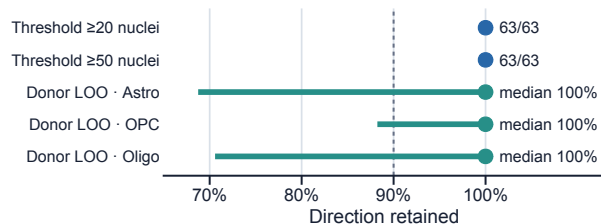


c

Direction robustness

Threshold sensitivity and donor leave-one-out.

Points: mean or median; bars: range across axes.



d

Captured-nucleus composition context

Eight of 42 primary CLR contrasts pass global FDR < 0.05; model-based 95% CI.

